Phase 0 — Environment Setup and Validation

Harrington Capital plc | Infrastructure Engineer Simulation

Overview

Before any technical work begins you must set up and validate your full working environment. You cannot progress to Phase 1 until every tool below is installed, configured, and producing verified output.

BUILD → VERIFY → SUBMIT on every tool.

Step 1 — GitHub Repository

Build

1. Create a GitHub account at https://github.com/signup if you do not have one.
2. Fork or clone the simulation repository: `https://github.com/alain-Sortnext/harrington-infrastructure-operations`
3. Clone it locally:

```
git clone https://github.com/YOUR-USERNAME/harrington-infrastructure-operations.git
cd harrington-infrastructure-operations
```

1. Install GitHub Desktop if you prefer a GUI: https://desktop.github.com

Verify

```
git log --oneline -3
git remote -v
```

Submit

- Your repository URL: `https://github.com/YOUR-USERNAME/harrington-infrastructure-operations`

- First commit SHA (copy from `git log --oneline -1`)
- Screenshot of the repository folder structure open in VS Code

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Step 2 — Microsoft Azure Free Account

Build

1. Go to https://azure.microsoft.com/en-gb/free/
2. Create a free account (requires credit card for identity — no charge for free tier).
3. Install Azure CLI:
4. Windows: https://aka.ms/installazurecliwindows
5. Linux/Mac: `curl -sL https://aka.ms/InstallAzureCLIDeb | sudo bash`
6. Log in:

```
az login
```

1. Create a resource group for the project:

```
az group create \
  --name rg-harrington-prod \
  --location uksouth \
  --tags Project=harrington-infrastructure-operations Environment=prod ManagedBy=CLI
```

Verify

```
az account show
az group show --name rg-harrington-prod
```

Submit

- Screenshot of Azure Portal showing `rg-harrington-prod` resource group
- Copy of `az account show` output (shows subscription ID and account name)

Step 3 — AWS Free Tier Account

Build

1. Go to https://aws.amazon.com/free/
2. Create a free account.
3. Install AWS CLI v2:
4. https://docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html
5. Configure credentials:

```
aws configure
# AWS Access Key ID: [paste from IAM console]
# AWS Secret Access Key: [paste from IAM console]
# Default region name: eu-west-2
# Default output format: json
```

Verify

```
aws sts get-caller-identity
```

Expected output:

```
{
    "UserId": "AIDA...",
    "Account": "123456789012",
    "Arn": "arn:aws:iam::123456789012:user/your-username"
}
```

Submit

- Full output of `aws sts get-caller-identity` pasted into your submission

Step 4 — Windows Server 2022 in VirtualBox

Build

1. Download VirtualBox: https://www.virtualbox.org/wiki/Downloads
2. Download Windows Server 2022 Evaluation (180-day free): https://www.microsoft.com/en-us/evalcenter/evaluate-windows-server-2022
3. Create a new VM in VirtualBox:
4. Name: `HC-DC01`
5. Type: Microsoft Windows | Version: Windows 2022 (64-bit)
6. RAM: minimum 4096 MB (4 GB)
7. CPU: 2 cores
8. Disk: 60 GB (dynamically allocated)
9. Mount the ISO and install Windows Server 2022 Datacenter (Desktop Experience).
10. After installation, rename the server:

```
Rename-Computer -NewName "HC-DC01" -Restart
```

Verify

Take a screenshot showing: - Windows Server 2022 desktop - Server name `HC-DC01` visible in System Properties or PowerShell:

```
$env:COMPUTERNAME
```

Submit

- Screenshot of HC-DC01 desktop showing server name

Step 5 — Ubuntu Server 22.04 LTS in VirtualBox

Build

1. Download Ubuntu Server 22.04 LTS ISO: https://ubuntu.com/download/server
2. Create a new VM in VirtualBox:
3. Name: `HC-APP01`
4. Type: Linux | Version: Ubuntu (64-bit)
5. RAM: 2048 MB
6. CPU: 2 cores
7. Disk: 40 GB
8. During installation, enable OpenSSH server when prompted.
9. After installation, set a static IP or note the DHCP address.
10. Test SSH from your host machine:

```
ssh hcadmin@<HC-APP01-IP>
```

Verify

```
uname -a
hostname
ip addr show
```

Submit

- Screenshot of SSH session showing hostname `HC-APP01` and Ubuntu version

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Step 6 — Docker Desktop

Build

1. Download Docker Desktop:
2. Windows/Mac: https://www.docker.com/products/docker-desktop/
3. Linux: https://docs.docker.com/engine/install/ubuntu/
4. Install and start Docker Desktop.
5. Run the verification container:

```
docker run hello-world
```

Verify

The output must contain:

```
Hello from Docker!
This message shows that your installation appears to be working correctly.
```

Submit

- Full output of `docker run hello-world` pasted into submission

Step 7 — Terraform

Build

1. Install Terraform:
2. Windows (winget): `winget install HashiCorp.Terraform`
3. Linux: https://developer.hashicorp.com/terraform/install
4. Mac (brew): `brew tap hashicorp/tap && brew install hashicorp/tap/terraform`

Verify

```
terraform version
```

Expected: `Terraform v1.6.x` or higher.

Submit

- Output of `terraform version` pasted into submission

Step 8 — PowerShell 7

Build

1. Check current version:

```
$PSVersionTable
```

1. If version is below 7.0, install PowerShell 7: https://aka.ms/install-powershell

Verify

```
$PSVersionTable
```

Must show `PSVersion 7.x.x`.

Submit

- Output of `$PSVersionTable` pasted into submission

Step 9 — Visual Studio Code

Build

1. Download VS Code: https://code.visualstudio.com/download
2. Install the following extensions (Ctrl+Shift+X):
3. `hashicorp.terraform` (Terraform)
4. `ms-vscode.powershell` (PowerShell)
5. `ms-azuretools.vscode-docker` (Docker)
6. `redhat.vscode-yaml` (YAML)
7. `yzhang.markdown-all-in-one` (Markdown)
8. Open the cloned repository folder in VS Code: `File → Open Folder → harrington-infrastructure-operations`

Verify

Take a screenshot showing VS Code with the repository folder open and at least 3 extensions visible in the Extensions panel.

Submit

- Screenshot of VS Code with extensions installed and repository open

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Step 10 — Prometheus and Grafana

Build

On HC-APP01 or HC-MON01 (Ubuntu):

Install Prometheus:

```
sudo apt update
sudo apt install -y prometheus
sudo systemctl enable prometheus
sudo systemctl start prometheus
```

Install Grafana:

```
sudo apt install -y apt-transport-https software-properties-common
wget -q -O - https://packages.grafana.com/gpg.key | sudo apt-key add -
echo "deb https://packages.grafana.com/oss/deb stable main" | sudo tee /etc/apt/sources.list.d/grafa
sudo apt update
sudo apt install -y grafana
sudo systemctl enable grafana-server
sudo systemctl start grafana-server
```

Verify

- Prometheus: open `http://<server-ip>:9090` — Status menu should show.
- Grafana: open `http://<server-ip>:3000` — Login page appears (admin / admin).

Submit

- Screenshot of Prometheus UI at `/targets` page
- Screenshot of Grafana login page

Step 11 — ITSM Tool (GLPI)

Build

Option A — GLPI (recommended for realism):

```
# On Ubuntu HC-APP01
sudo apt install -y apache2 php php-mysql php-curl php-gd php-mbstring \
    php-xml php-xmlrpc php-zip mariadb-server

# Download GLPI
wget https://github.com/glpi-project/glpi/releases/download/10.0.14/glpi-10.0.14.tgz
tar -xzf glpi-10.0.14.tgz -C /var/www/html/
sudo chown -R www-data:www-data /var/www/html/glpi
```

Follow the web installer at `http://<server-ip>/glpi`

Option B — Jira Service Management Free:

1. Go to https://www.atlassian.com/software/jira/service-management
2. Sign up for the free tier (up to 3 agents).
3. Create a project named: `Harrington Capital IT Support`

Verify

Create a test incident ticket: - Title: `TEST-001 — Environment validation ticket` - Priority: P3 - Category: Infrastructure

Note the ticket number.

Submit

- Screenshot of the ITSM tool showing the test ticket
- Ticket ID / number (e.g. `INC-20260601-001` or GLPI ticket #1)

Step 12 — Networking Tools

Build

Install the following:

```
# Ubuntu
sudo apt install -y wireshark nmap dnsutils net-tools traceroute

# Windows (winget)
winget install WiresharkFoundation.Wireshark
```

Download Draw.io Desktop: https://app.diagrams.net or https://github.com/jgraph/drawio-desktop/releases

Install Windows Terminal (Windows): https://aka.ms/terminal

Verify

```
# Linux
dig harrington.local @<HC-DC01-IP>
nslookup harrington.local <HC-DC01-IP>
ping -c 4 <HC-APP01-IP>
traceroute <HC-APP01-IP>
```

Submit

- Output of `dig` or `nslookup` query
- Output of `ping` and `traceroute`

Phase 0 Complete Checklist

Before submitting, confirm every item below:

#	Item	Evidence Required
1	GitHub repository URL	URL in submission
2	First commit SHA	40-character SHA
3	Azure Resource Group screenshot	Portal screenshot
4	AWS identity verification	`aws sts get-caller-identity` output
5	Windows Server HC-DC01 screenshot	Desktop + server name visible
6	Ubuntu HC-APP01 SSH screenshot	Terminal showing hostname
7	`docker run hello-world` output	Full console output
8	`terraform version` output	Console output
9	`$PSVersionTable` output	Console output
10	VS Code with extensions screenshot	Extensions panel visible
11	Prometheus UI screenshot	`/targets` page
12	Grafana login page screenshot	Login page at port 3000
13	ITSM test ticket screenshot	Ticket number visible

All 13 items must be submitted to unlock Phase 1.